Final Project - Step 3

PSTAT100: Data Science Concepts and Analysis

Abstract

This project investigates how study habits, mental health, and lifestyle factors influence academic performance among students using a simulated dataset. Key variables include daily study hours, self-reported mental health, sleep duration, physical activity, social media use, and Netflix viewing time. A multiple linear regression model was used to quantify the effect of these factors on final exam scores. The analysis found that study time and mental health were the strongest positive predictors of exam performance, while time spent on social media and Netflix had significant negative effects. Sleep and physical activity had smaller but positive associations.

Introduction

The data used is a simulated dataset from Kaggle, that explores how lifestyle habits affect academic performance in students. Key variables include:

- `study_hours_per_day`: Average daily study time (hours).
- `mental_health_rating`: Self-reported mental health score (scale 1-10).
- `sleep_hours`: Average nightly sleep duration (hours).
- `exercise_frequency`: Hours per week of physical activity.
- `social_media_hours`: Daily time on social media (hours).
- `netflix_hours`: Daily time watching Netflix (hours).
- `exam_score`: Final exam score (0-100)

We investigate three research questions:

1. **Study Time**: How does average daily study time relate to final exam scores?
2. **Mental Health**: To what extend does self-reported mental health score predict exam performance?
3. **Lifestyle**: Which non-academic habits (sleep, physical activity, social media, Netflix) are most strongly associated with exam scores?

With the hypotheses:

- H_1 (Study Time): Higher study time (>2 hours) is associated with higher exam scores.
- H_2 (Mental Health): Better mental health is associated with higher exam scores.
- H_3 (Lifestyle): Sleep and physical activity are positively associated with exam scores, whereas social media and Netflix usage are negatively associated.

Data Processing

Missing Data Assessment

Table 1: Number of Missing Values by Numeric Variable

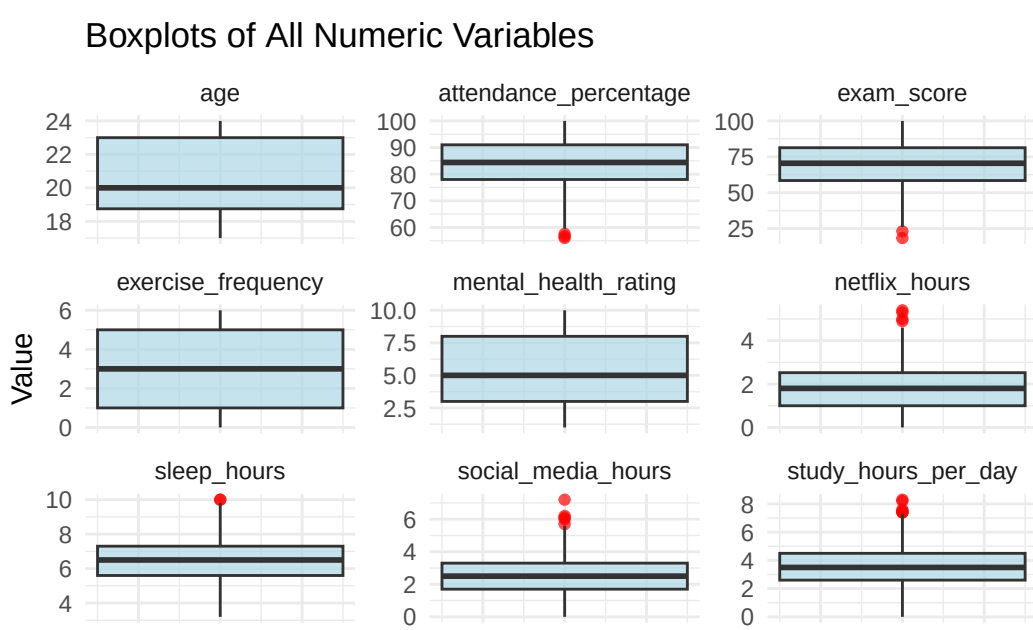
Variable	Number of Missing Values
study_hours_per_day	0
mental_health_rating	0
sleep_hours	0
exercise_frequency	0
social_media_hours	0
netflix_hours	0
exam_score	0

No missing values were detected for any variable.

Outlier Detection and Assessment

Table 2: Observations with $|Z| > 3$

Row	Variable	Z-Score	Original Value
736	social_media_hours	4.00	7.2
557	netflix_hours	3.33	5.4
823	netflix_hours	3.24	5.3
456	study_hours_per_day	3.23	8.3
798	study_hours_per_day	3.17	8.2
146	social_media_hours	3.15	6.2
362	social_media_hours	3.07	6.1
266	exam_score	-3.03	18.4



While several observations exceed ± 3 standard deviations, these appear to represent legitimate extreme values rather than data entry errors. Removing them could eliminate valid information about the tails of the distributions.

Correlation Analysis

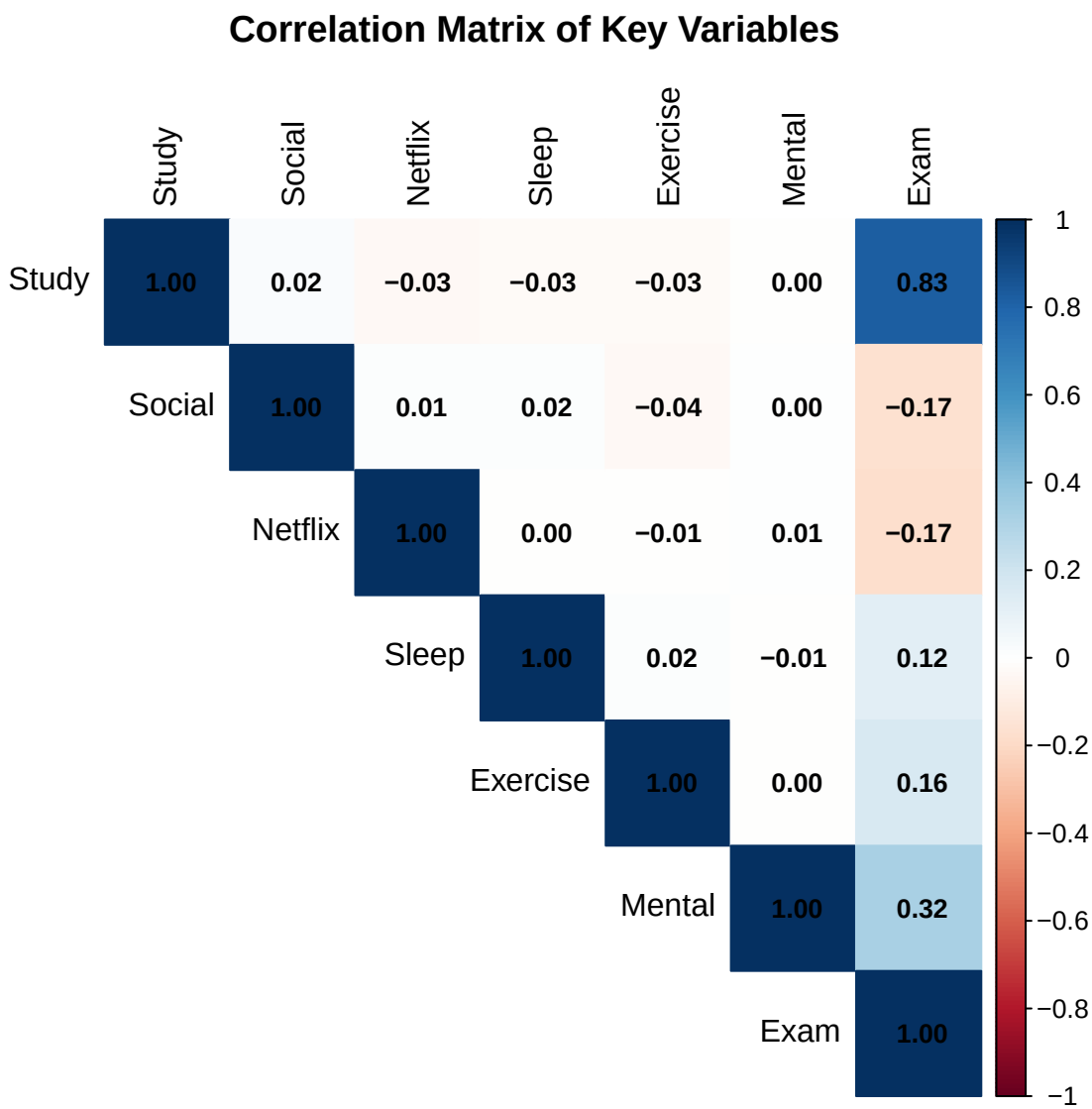


Figure 1: Correlation Plot shows some strong positive correlation between a couple of variables and exam scores.

The correlation analysis reveals strong, positive correlations between exam scores and both study hours ($r = 0.83$) and mental health ($r = 0.32$), providing initual support for hypotheses H_1 and H_2. Negative correlations with social media and Netflix usage ($r = -0.17$) support hypothesis H_3.

Modeling Process

Model Selection and Justification

Multiple linear regression was selected as the primary analytical approach because:

1. **Continuous Outcome**: Exam scores are measured on a continuous scale (0-100)
2. **Multiple Predictors**: We need to assess simultaneous effects of six predictors
3. **Hypothesis Testing**: Linear regression allows direct testing of our hypotheses
4. **Interpretability**: Coefficients provide clear, actionable insights about effect sizes

Model Fitting

Table 3: Multiple Linear Regression Results

Variable	Coefficient	Std. Error	t-value	p-value	95% CI Lower	95% CI Upper	Sig.
(Intercept)	18.100	1.213	14.92	< 0.001	15.719	20.481	***
study_hours_per_day	9.598	0.119	80.91	< 0.001	9.365	9.831	***
mental_health_rating	1.940	0.061	31.75	< 0.001	1.820	2.060	***
sleep_hours	2.020	0.142	14.23	< 0.001	1.741	2.298	***
exercise_frequency	1.448	0.086	16.83	< 0.001	1.279	1.617	***
social_media_hours	-2.574	0.149	-17.32	< 0.001	-2.865	-2.282	***
netflix_hours	-2.279	0.162	-14.07	< 0.001	-2.597	-1.961	***

Table 4: Model Performance Statistic

R^2	Adj. R^2	RMSE	F-statistic	F p-value	df	Residual df
0.8946	0.894	5.5	1404.7	< 0.001	6	993

Assumption Validation

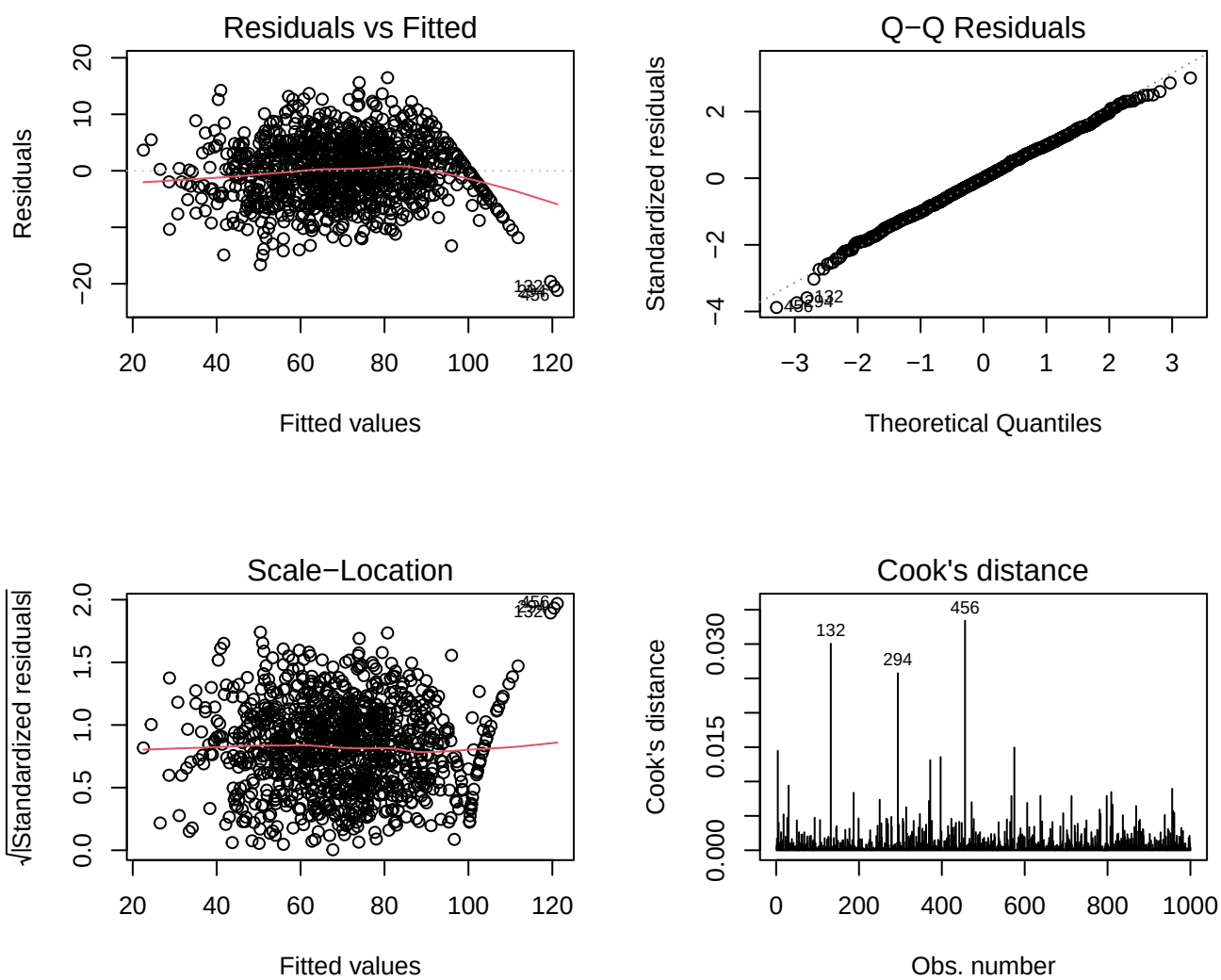


Figure 2: Diagnostic plots confirm that linear regression assumptions are adequately satisfied for valid inference.

Table 5: Variance Inflation Factors (VIF)

Variable	VIF
study_hours_per_day	1.0030
mental_health_rating	1.0001
sleep_hours	1.0016
exercise_frequency	1.0026
social_media_hours	1.0023
netflix_hours	1.0012

```
Diagnostic Test Results:

Shapiro-Wilk (Normality): p = 0.1384

Durbin-Watson (Independence): p = 0.532

VIF Assessment: All values < 5
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Assumption Assessment:

- **Linearity & Homoscedasticity**: Residuals vs. fitted plot shows random scatter around zero with no clear patterns.
- **Normality**: Q-Q plot shows good alignment with theoretical quantiles; Shapiro-Wilk test ($p = 0.1384$) supports normality.
- **Independence**: Durbin-Watson test ($p = 0.532$) indicates no autocorrelation.
- **Multicollinearity**: All VIF values ≈ 1, indicating no concerning collinearity.
- **Influential Points**: The Cook's distance graph shows a few individual observations that have a high influence on the regression model (456, 132, 294). Nothing appears extreme to invalidate or remove from the model.

Since the assumptions for linear regression are met, we may proceed with confidence regarding coefficient interpretations and statistical inference.

Results

Overall Model Performance

The multiple linear regression model demonstrates excellent predictive performance:

- $\mathbf{R^2 = 0.8964}$: The model explains 89.64% of variance in exam scores.
- **Adjusted** $\mathbf{R^2 = 0.894}$: Accounts for multiple predictors. Very similar to R^2, indicating a robust fit.
- $\mathbf{F(6, 993) = 1404.7,\ p < 0.001}$: Highly significant overall model.
- $\mathbf{RMSE = 5.5}$: Average prediction error of ~ 5.5 points.

Hypothesis Testing Results

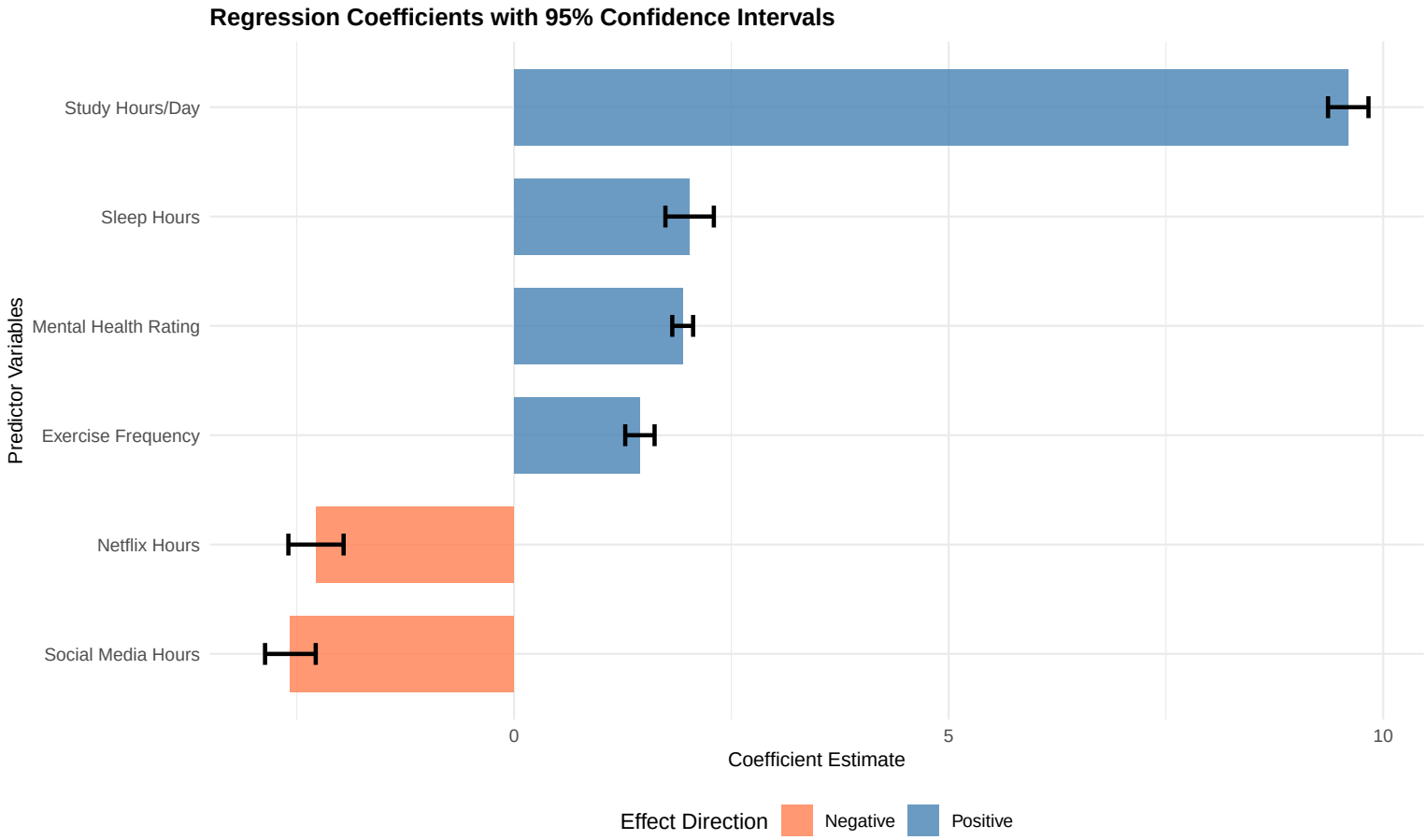

Figure 3: Coefficient plot demonstrates the relative importance and direction of each predictor, with study time showing the strongest positive effect.

Detailed Hypothesis Results:

H_1 - Study Time Effect:

- **Coefficient**: 9.598 (95% CI: $9.365, 9.831$)
- **Interpretation**: Each additional hour of daily study increases exam scores by ~ 9.6 points
- **Statistical Significance**: $p < 2 \times 10^{-16}$ (highly significant)
- **Conclusion**: H_1 is strongly supported

H_2 - Mental Health Effect:

- **Coefficient**: 1.94 (95% CI: $1.82, 2.06$)
- **Interpretation**: Each 1 point increase in mental health rating increases exam scores by 1.94 points.
- **Statistical Significance**: $p < 2 \times 10^{-16}$ (highly significant)
- **Conclusion**: H_2 is strongly supported

H_3 - Lifestyle Factors:

- **Sleep Hours**: $\beta = 2.02 \quad (p < 2 \times 10^{-16})$ - Positive as predicted
- **Exercise Frequency**: $\beta = 1.448 \quad (p < 2 \times 10^{-16})$ - Positive as predicted
- **Social Media Hours**: $\beta = -2.574 \quad (p < 2 \times 10^{-16})$ - Negative as predicted
- **Netflix Hours**: $\beta = -2.279 \quad (p < 2 \times 10^{-16})$ - Negative as predicted
- **Conclusion**: H_3 is fully supported

Interpretation

Key Findings and Implications

Academic Factors

Study Time: The analysis reveals study time as the most powerful predictor of exam performance. With a coefficient of 9.598, each additional hour of daily study translates to nearly 10 points higher on exams. This effect is both statistically significant ($p < 2 \times 10^{-16}$) and practically meaningful. A student studying 4 hours daily versus 2 hours would score approximately 20 points higher, equivalent to moving from a C+ to an A-.

Mental Health: Mental health is the third strongest predictor ($\beta = 1.94$), highlighting the crucial role of psychological well-being in academic success. A student with excellent mental health (rating 9/10) would score about 15 points higher than someone with poor mental health (rating 1/10), demonstrating that academic interventions must address mental health alongside study skills.

Lifestyle Factors

Sleep Duration ($\beta = 2.02$): Each additional hour of sleep contributes ~ 2 points to exam scores. Students getting 8 hours vs. 6 hours of sleep would score ~ 4 points higher, supporting the importance of adequate rest for cognitive function.

Exercise Frequency ($\beta = 1.448$): Each additional hour of weekly exercise adds ~ 1.5 points to exam scores. Individually, exercise frequency is a modest predictor. Although, it represents the cumulative benefit of physical activity on cognitive performance and stress management.

Screen Time: Both social media ($\beta = -2.574$) and Netflix usage ($\beta = -2.279$) show significant negative associations. A student spending 4 hours daily on social media versus 2 hours would score approximately 5 points lower, demonstrating the costs of excessive recreational screen time.

Model Performance and Reliability

The model's exceptional performance ($R^2 = 0.8946$) indicates that these six variables capture nearly 90% of the variation in exam scores, suggesting that the model has identified the key drivers of academic performance. The low RMSE (5.5) means predictions are typically accurate within ± 5.5 points, making the model highly reliable for practical applications.

Visualization and Communication

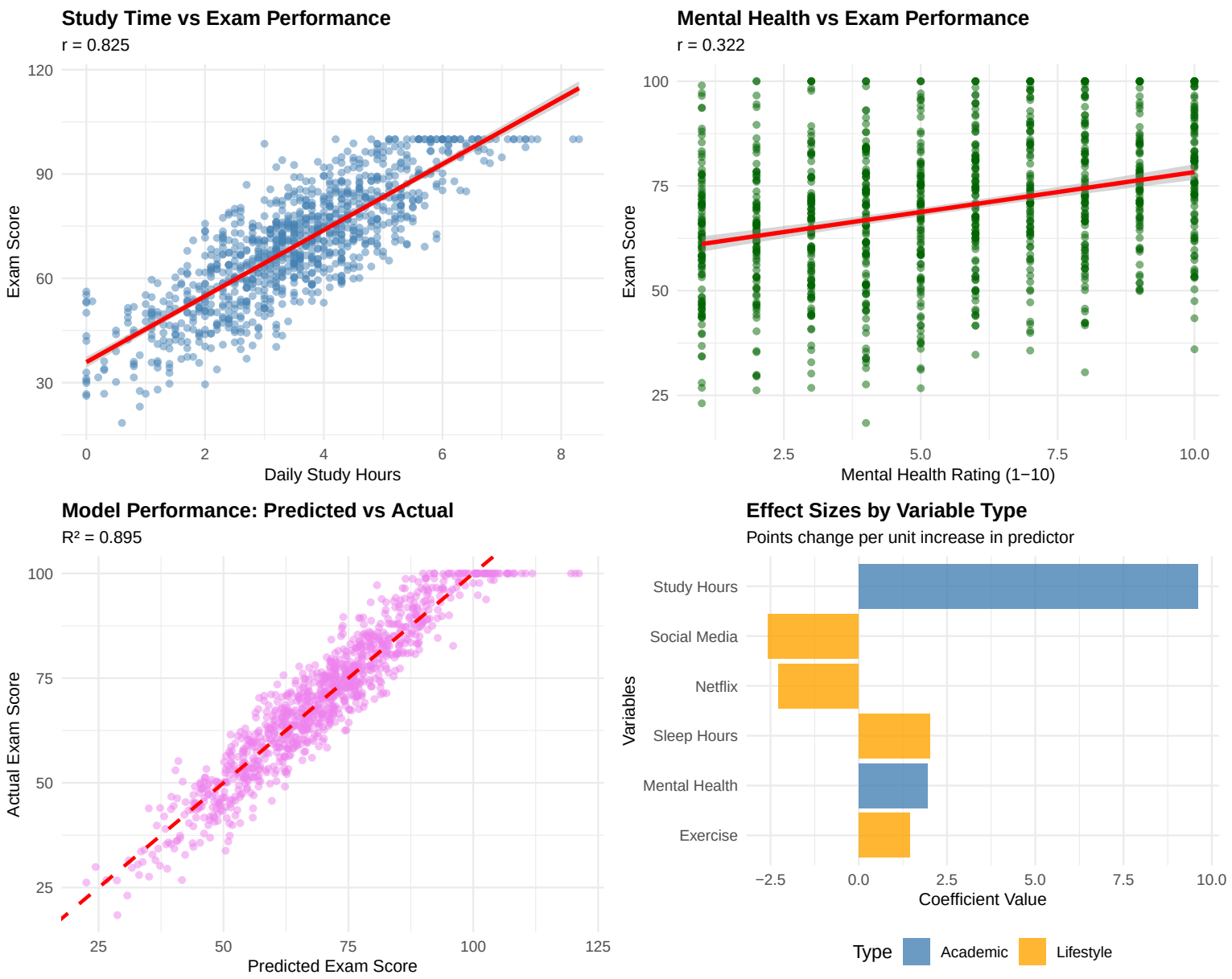

Figure 4: Comprehensive visualizations demonstrates model performance, key relationships, and residual patterns for complete analysis.

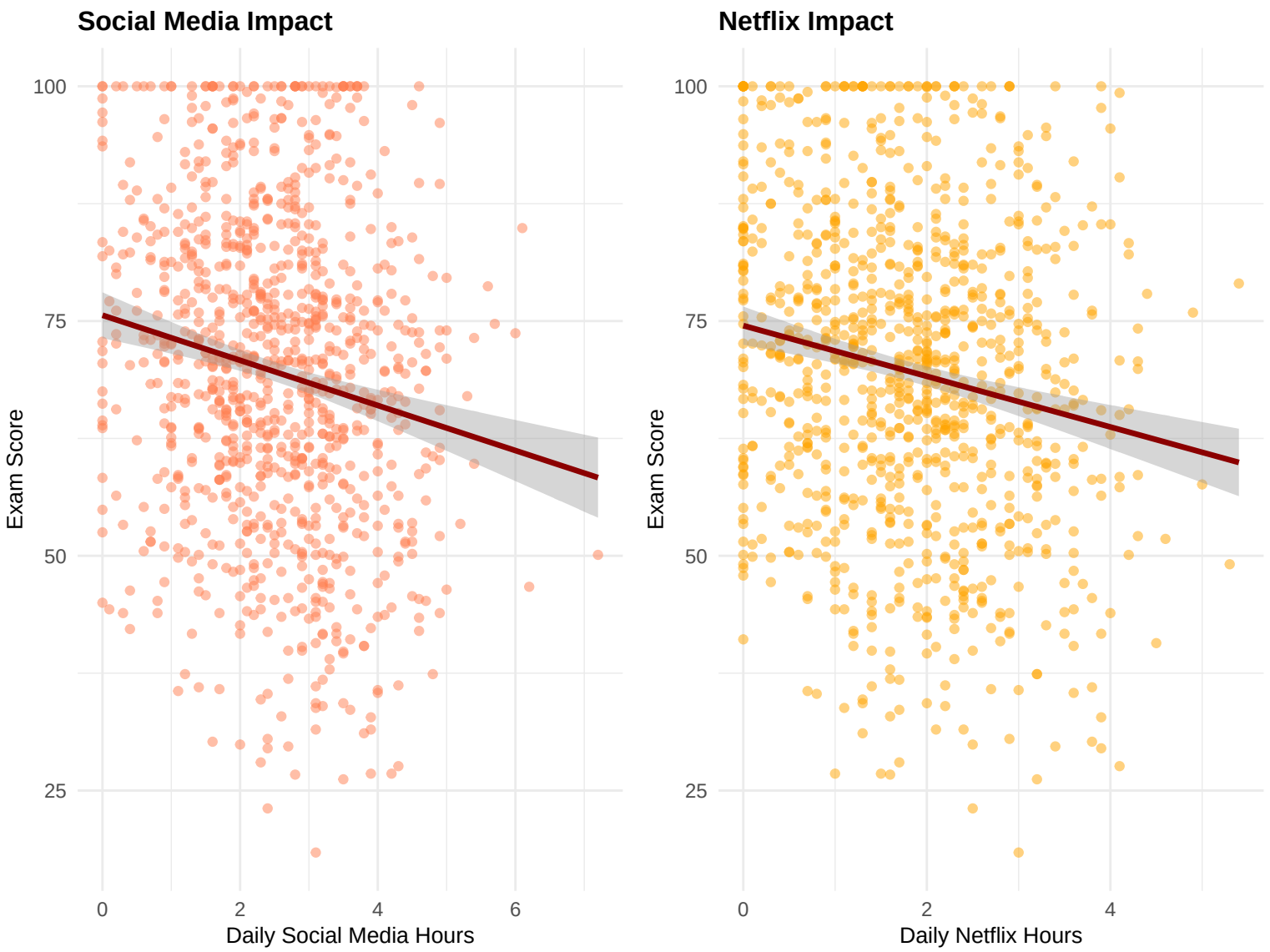

Figure 5: Lifestyle factors show clear but moderate associations with exam performance, with screen time variables demonstrating expected negative relationships.

The visualizations confirm strong linear relationships for academic predictors while revealing more moderate but consistent patterns for lifestyle factors. The predicted vs. actual plot demonstrates excellent model fit with minimal bias across the score range.

Conclusion and Recommendations

Summary of Key Insights

All the variables are statistically significant in their effect on exam scores, however they each have varying influence of their effect. Students' study hours per day had the greatest positive impact on exam scores, with mental health rating having the third greatest impact on exam scores. Sleep and exercise had a slightly positive impact on exam scores. Social media hours and Netflix hours both has a slightly negative impact on exam scores. These results were as expected and support each of our hypotheses. H_1 outlined that higher study time (>2 hours) would be associated with higher exam scores, which we found to be true. H_2 outlined that better mental health scores would be associated with higher exam scores, which we found to be true. H_3 outlined that sleep and physical activity are positively associated with exam scores, whereas social media and Netflix are negatively associated with exam scores, which we also found to be true.

Recommendations

For Individual Students:

- Optimize study habits: Establish consistent daily study routines, targeting 3-4 hours of study time.
- Prioritize mental health: Develop stress management strategies and seek counseling or mental health support when experiencing difficulties.
- Implement healthy lifestyle practices: Maintain 7-8 hours of sleep nightly for optimal cognitive function. Incorporate regular physical activity, aiming for 3-5 hours weekly. Also, limit recreational screen time to < 2 hours daily for social media and Netflix.

For Educational Institutions:

- Academic support services: Develop study skills workshops focusing on time management and effective learning strategies. Also, create quiet, well-equipped study spaces accessible to all students.
- Mental health infrastructure: Expand counseling services and reduce wait times for appointments, and provide mental health literacy programs and stress management workshops.
- Holistic wellness programs: Provide fitness facilities and recreational activities, and develop policies that support work-life balance

Limitations and Future Directions

The dataset is labeled as "simulated yet realistic dataset" meaning all the data was randomly generated. Thus, we are limited in our interpretation of this data because while it is intended to be realistic, it is not derived from real students' data. Moreover, variables like "internet quality" and "diet quality" were excluded from the model due to their subjective and self-assessed nature, with responses like "good," "average," and "poor" having different interpretations among students.

Potential future directions include using a real-world dataset rather than simulated data to eliminate the limitations associated with synthetic responses. This would improve the generalization and practical relevance of the findings. Additionally, future research could incorporate additional predictors that were excluded here due to the subjective nature of the survey responses (e.g., "diet quality" or "internet quality"). To address this, future research could focus on refining survey design by using more objective wording, clearer response categories, or standardized scales to reduce ambiguity and improve data quality.